

Quantifying a social problem - The dutch housing crisis

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2026-06-17

Title Page

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Include the tutorial group number

Include your tutorial lecturer's name

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

Include the following:

- Why is this relevant?
- ...

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

midwest is an example dataset included in the tidyverse package

2.2 Provide a short summary of the dataset(s)

Linking datasets

The datasets were linked using the variables "Regio.s" and "Perioden". These variables are present in both datasets and identify the same region and year.

A left join was chosen because the household dataset was used as the primary dataset. Housing stock information was added for all matching regions and years

In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

```
inline_code = TRUE
```

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

For the sake of this example, I will continue with the assignment...

Part 3 - Quantifying

3.1 Data cleaning

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

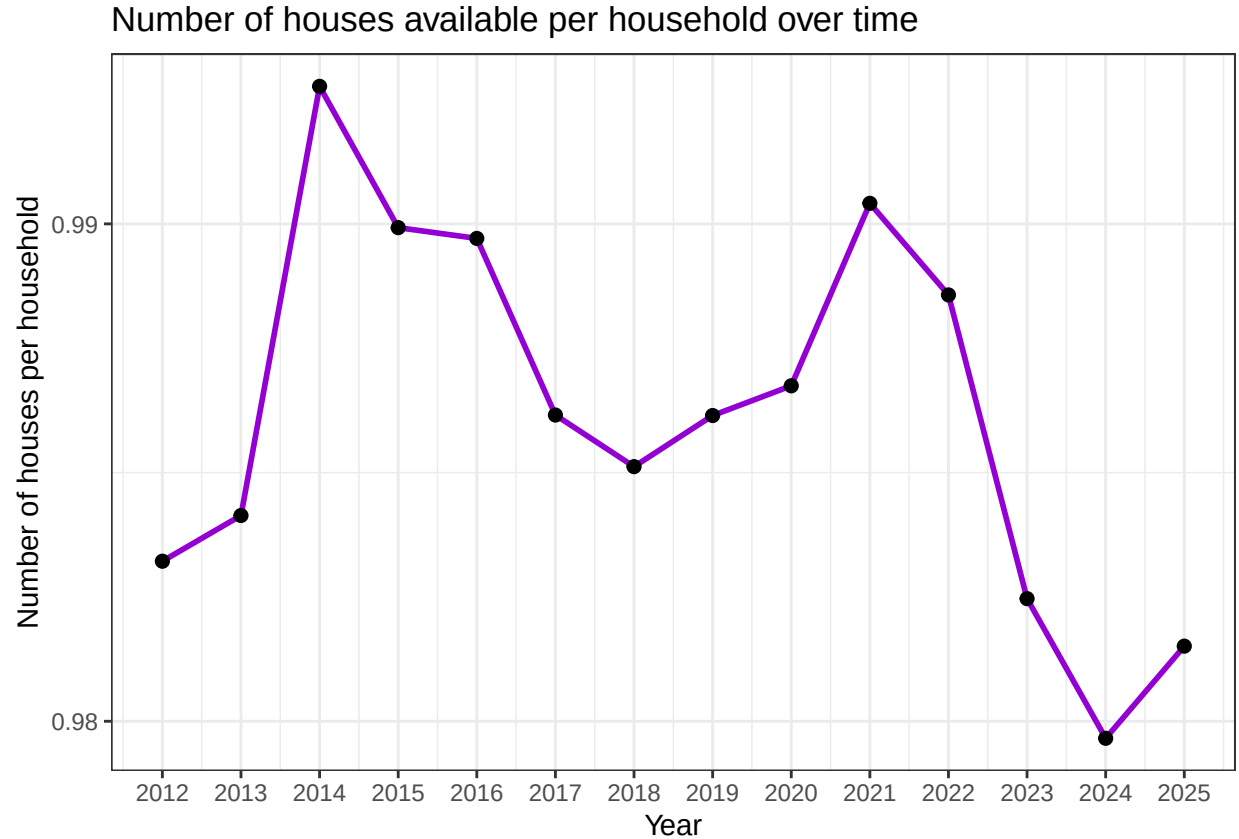
Variabelen hernoemen

Variable 1

Variable 2 Als tweede variabelen gaan wij een percentuele verdeling van alle woningen door heel nederland. Dus dan moeten wij het totale aantal woningen delen door de aantal beschikbare woningen per regio. Zo krijgen wij dus te zien hoeveel % van het totale aantal staat in een provincie

3.3 Visualize temporal variation

To examine changes in housing availability over time, a new variable was created by dividing the total housing stock by the total number of private households in the Netherlands. This variable, "woningen_per_huishouden)", provides an indication of the balance between the supply of housing and the number of households. The data was filtered to include only observations for the Netherlands, and records with missing values for the housing availability indicator were excluded. A line plot was then used to visualise how the ratio evolved over time.



The figure shows the number of houses available per household in the Netherlands over time. This indicator is relevant to the housing shortage because it relates the housing stock directly to the number of households that require housing. A declining ratio would suggest that household growth is outpacing the expansion of the housing stock, potentially increasing pressure on the housing market. Conversely, a stable or increasing ratio would indicate that housing supply is keeping pace with household formation. The trend therefore provides insight into the development of housing availability in relation to demand.

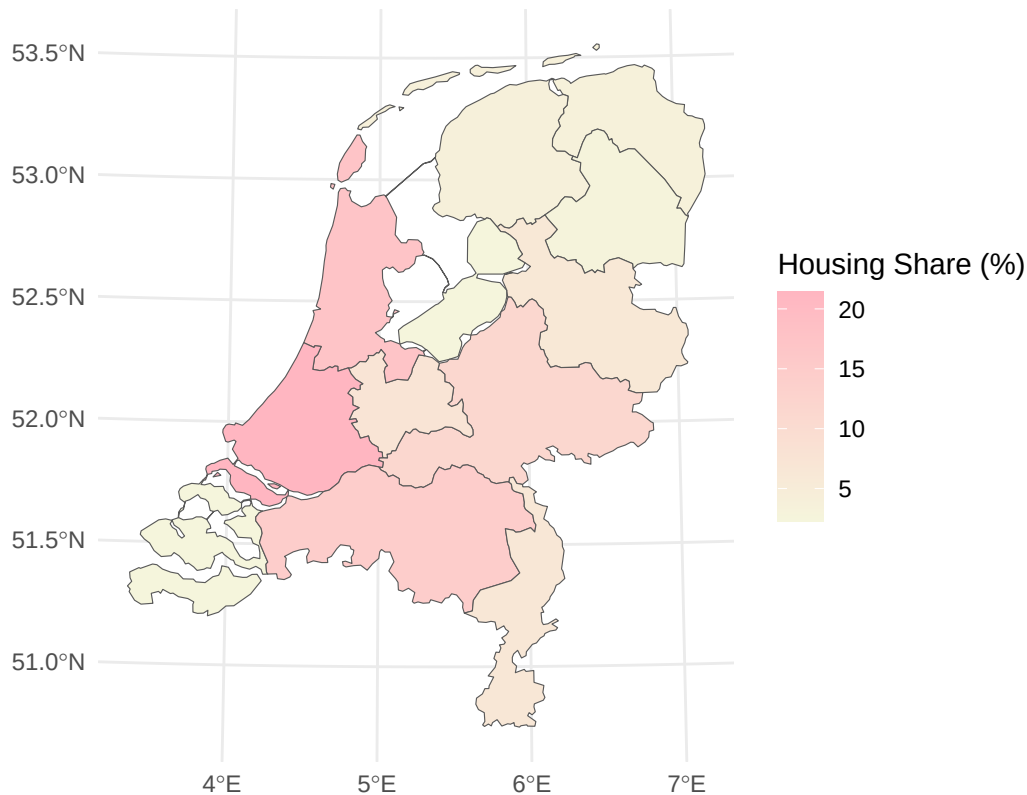
3.4 Visualize spatial variation

To analyze the spatial variation in the Dutch housing market, the share of the national housing stock in each province was calculated for 2025. This was done by dividing the housing stock of each province by the total housing stock of the Netherlands and expressing the result as a percentage.

The data was filtered to include only provinces and not the national total. The names of the provinces were cleaned to ensure compatibility with the CBS geographic shapefile, after which the housing data were connected to the provincial boundaries using a left-hand link.

A choropleth map was used because geographical differences in housing distribution are easier to identify on a map than in a conventional table or graph.

Proportional Distribution of Dutch Housing Stock by Province (2025)



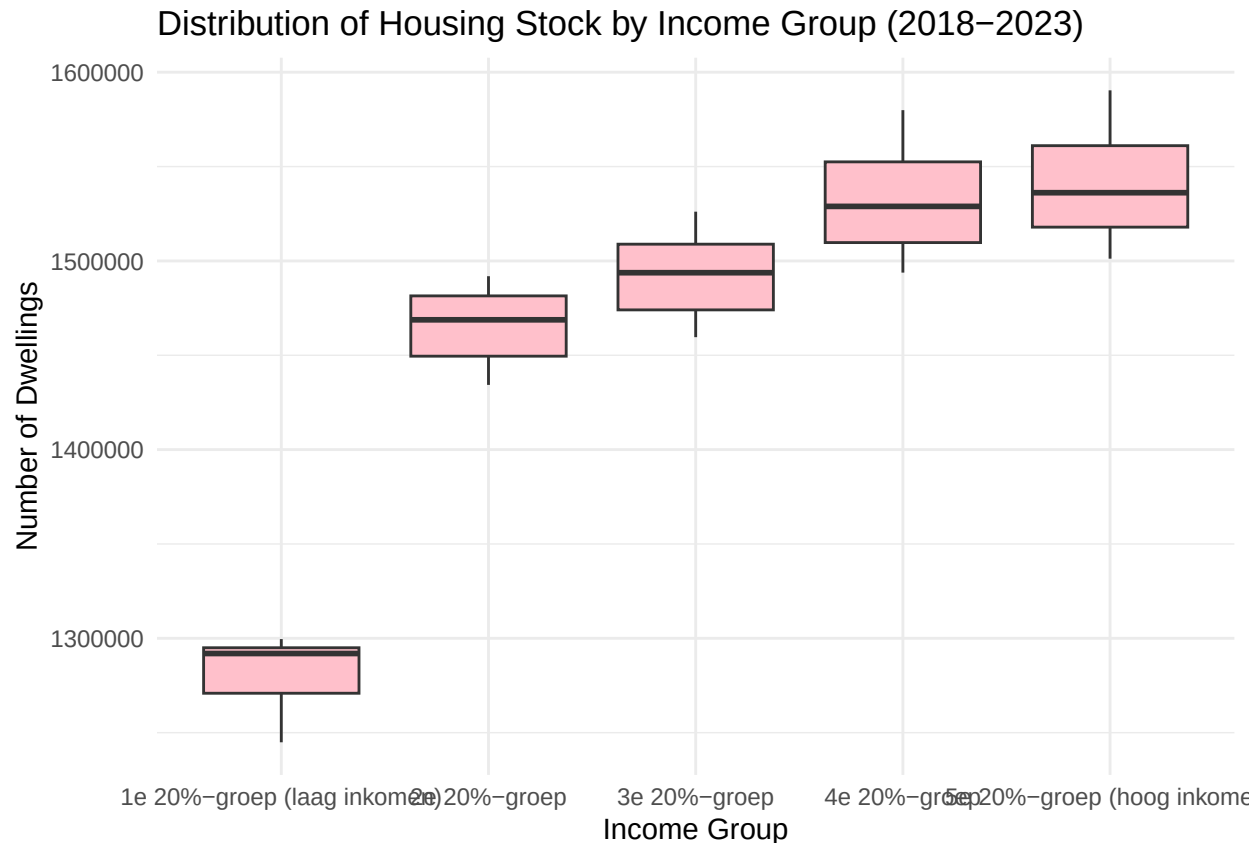
The map shows how the Dutch housing stock is distributed across provinces in 2025. Zuid-Holland, Noord-Holland, and Noord-Brabant have the largest share of the national housing stock, while provinces such as Zeeland, Drenthe, and Flevoland have the smallest shares.

This pattern reflects the concentration of population and economic activity in the western and southern parts of the Netherlands. The map is relevant to the Dutch housing shortage because it shows where the housing supply is geographically concentrated. Knowledge of the spatial distribution of the housing stock helps to identify regional imbalances between supply and demand in the housing market and provides insight into where additional housing construction is needed.

Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

To investigate variation across sub-populations, housing stock levels were compared across income groups. Households with unknown income were excluded from the analysis. A boxplot was used to visualise differences in the distribution of housing stock between income categories over time.



The boxplot shows differences in housing stock across income groups. Comparing the distributions allows us to assess whether certain income groups consistently account for a larger share of the housing stock. This is relevant to the housing shortage because housing availability may not be equally distributed across socioeconomic groups.

Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

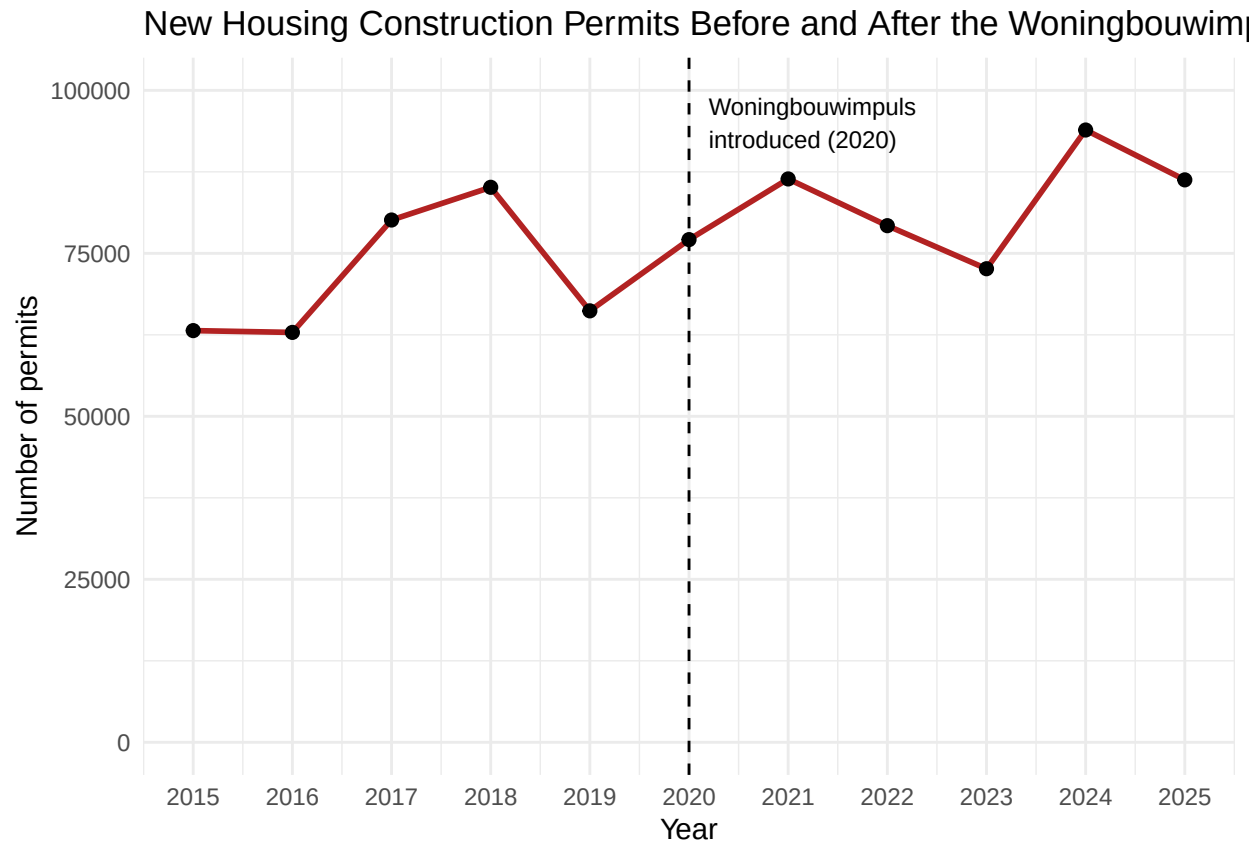
Analyze the relationship between two variables.

To examine whether a major housing policy intervention was associated with changes in housing development, an event analysis was conducted using data on permits issued for new housing construction in the Netherlands between 2015 and 2025.

The data was filtered to include only residential buildings (“Woning”) at the national level (“Nederland”). Incomplete observations for 2026 were excluded from the analysis because only partial monthly data were available, whereas the other observations represent complete calendar years. The year variable was converted to a numeric format to enable temporal visualisation in a line plot.

A line plot was used to visualise changes over time, while a vertical reference line marks the introduction of the Woningbouwimpuls in 2020.

```
## Warning: There was 1 warning in 'filter()'.
## i In argument: '!is.na(as.numeric(Perioden))'.
## Caused by warning:
## ! NAs introduced by coercion
```



The figure shows fluctuations in the number of permits issued for new housing construction between 2015 and 2025. Following the introduction of the Woningbouwimpuls in 2020, permit levels increased in some years but also experienced periods of decline. The highest number of permits was recorded in 2024. Although the trend does not show a clear and immediate structural change after 2020, the figure illustrates how housing development evolved during a period in which the Dutch government implemented policies aimed at increasing housing supply. This is relevant to the housing shortage because new housing construction plays an important role in expanding the available housing stock.

Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.